

Analyzing the Impact of Environmental Conditions on Driving Scene Difficulty Using BDD100K

Team Information

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Dataset Overview

We propose to use the BDD100K dataset from Berkeley DeepDrive. BDD100K is a large-scale driving dataset containing 100,000 videos collected under diverse driving conditions. It includes environmental attributes such as weather, time of day, and scene type, along with annotations for perception tasks like object detection. This dataset is suitable for our project because it is large, diverse, well-labeled, and relevant to autonomous driving. Its variety in environmental conditions makes it useful for studying how scene context affects visual perception difficulty.

Research Problem and Motivation

Our project studies how environmental conditions affect the difficulty of visual perception in driving scenes. We want to examine whether conditions such as nighttime, rainy weather, or certain scene types are associated with higher detection difficulty. This matters because perception models in autonomous driving often perform unevenly across different environments, and understanding these patterns may help improve robustness and reliability. Our main question is: Can environmental and scene-related features help identify difficult driving scenes for visual perception tasks?

Preliminary Approach

We will preprocess a subset of BDD100K and extract structured features from scene metadata and object annotations. These features may include weather, time of day, scene category, image brightness, object counts, average bounding box size, and the proportion of small objects. We will define scene difficulty using indicators such as low visibility, high object density, or many small objects, and formulate the task as easy vs. hard scene classification. As baseline models, we will use logistic regression and decision trees, and we may further explore random forests or gradient boosting. We will also perform exploratory data analysis to compare scene difficulty across conditions.

Evaluation Plan

We will evaluate our methods using accuracy, precision, recall, and F1-score. Because the easy/hard classes may be imbalanced, we will focus especially on F1-score. We will consider the project successful if the models can distinguish difficult scenes from easier ones and if our analysis reveals clear relationships between environmental factors and scene difficulty.